

LAKE VYRNWY, United Kingdom

WMO: 034100

Lat: 52.7572N

Lon: 3.4653W

Elev: 1178

StdP: 14.08

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.2	27.3	20.9	16.3	29.4	23.1	18.1	30.0	53.0	42.6	43.6	44.7	8.2	70	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	72.7	63.5	69.1	60.9	66.1	59.2	64.3	69.7	62.1	66.9	60.5	64.5	7.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.3	87.8	66.2	60.3	81.7	63.6	58.8	77.3	61.8	30.0	69.8	28.4	67.1	27.3	64.7	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
34.1	28.5	24.3	22.2	76.4	3.6	4.1	19.7	79.3	17.6	81.8	15.6	84.1	13.0	87.1		
			21.6	68.0	3.4	3.8	19.1	70.8	17.1	73.0	15.2	75.2	12.7	78.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.1	37.9	38.4	40.6	44.6	49.5	54.5	57.6	57.0	53.7	48.6	43.0	39.2
	DBStd	8.59	5.36	5.21	5.17	5.14	5.00	4.51	4.46	3.81	3.85	4.52	4.86	5.80
	HDD50	1883	375	324	294	174	70	8	0	0	10	80	213	335
	HDD65	6540	840	744	758	612	480	315	234	251	338	509	660	799
	CDD50	825	0	0	1	12	55	144	237	217	121	35	3	0
	CDD65	11	0	0	0	0	0	2	6	3	0	0	0	0
	CDH74	54	0	0	0	0	0	8	31	15	0	0	0	0
	CDH80	3	0	0	0	0	0	0	2	1	0	0	0	0
Wind	WSAvg	9.8	11.8	11.8	10.3	8.8	9.3	8.3	8.2	8.3	8.5	10.1	10.2	11.6
Precipitation	PrecAvg	53.3	6.3	5.0	3.7	3.3	3.5	3.2	3.3	3.3	3.7	5.5	5.8	6.6
	PrecMax	69.8	10.7	12.2	7.5	6.5	6.3	7.4	7.4	5.3	7.8	11.1	14.3	14.4
	PrecMin	37.6	0.4	0.9	0.8	0.6	1.3	0.6	1.2	0.6	0.2	1.1	2.7	1.3
	PrecStd	8.0	3.0	2.8	1.7	1.7	1.5	1.8	1.5	1.2	1.9	2.8	2.9	3.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.0	53.3	56.9	65.2	70.5	74.8	77.6	76.2	69.9	62.1	55.4	52.7
		MCWB	49.0	48.3	49.2	54.2	60.7	63.3	65.1	63.9	60.6	57.4	53.7	51.5
	2%	DB	49.0	49.4	52.5	59.6	65.6	69.7	73.7	69.6	65.2	58.6	53.6	51.1
		MCWB	47.5	47.3	47.5	51.1	57.3	61.4	64.9	61.1	59.5	56.3	52.3	50.0
	5%	DB	47.3	47.6	50.0	55.6	61.5	66.2	70.0	66.1	62.3	56.9	51.9	49.3
		MCWB	45.9	45.7	46.6	49.0	54.7	58.9	62.1	59.7	57.8	54.5	50.4	48.1
	10%	DB	45.7	45.9	47.9	52.3	58.1	62.9	66.1	63.4	60.1	55.2	50.2	47.5
		MCWB	44.3	44.0	45.2	47.3	52.4	56.8	59.8	58.6	56.5	53.0	48.6	46.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.1	49.9	50.9	55.1	61.8	67.3	73.0	65.6	63.7	58.9	54.5	51.9
		MCDB	50.7	51.6	53.9	62.0	68.2	71.0	73.8	71.2	66.6	59.8	55.2	52.3
	2%	WB	47.7	47.8	48.8	52.2	58.5	62.3	64.9	62.6	60.9	56.9	52.7	50.2
		MCDB	48.6	49.2	51.2	57.4	64.0	67.3	70.7	66.6	63.7	58.5	53.4	50.9
	5%	WB	46.3	46.2	47.3	50.2	55.7	59.8	62.4	61.1	59.0	55.2	50.8	48.4
		MCDB	47.2	47.3	49.4	54.1	60.1	64.5	68.8	64.7	61.4	56.7	51.7	49.2
	10%	WB	44.5	44.3	45.7	48.3	53.5	57.6	60.4	59.6	57.2	53.4	48.8	46.3
		MCDB	45.6	45.8	47.7	51.8	57.2	61.9	64.9	62.7	59.5	54.7	49.9	47.3
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.8	9.4	11.5	12.1	12.1	11.9	10.8	10.5	8.5	7.4	7.1
		MCDBR	8.8	9.7	12.6	16.3	17.2	18.0	18.5	14.9	13.8	10.0	8.5	8.7
		MCWBR	9.1	8.6	9.1	10.0	10.5	11.7	11.1	8.6	9.5	8.3	8.4	9.2
	5% WB	MCDBR	8.5	9.0	11.1	14.1	15.5	15.9	16.1	12.1	12.4	9.5	8.5	8.7
		MCWBR	9.2	8.6	9.0	9.4	10.3	11.5	10.8	8.4	9.8	8.7	8.8	9.3
Clear-Sky Solar Irradiance	taub	0.312	0.326	0.355	0.377	0.382	0.375	0.374	0.359	0.350	0.335	0.312	0.302	
	taud	2.331	2.386	2.312	2.272	2.297	2.341	2.347	2.404	2.430	2.439	2.392	2.298	
	Ebn at Noon	212	248	263	271	275	278	276	274	263	242	212	191	
	Edh at Noon	19	24	32	38	39	37	36	33	28	23	18	17	
All-Sky Solar Radiation	RadAvg	222	429	788	1213	1516	1610	1507	1233	920	515	265	169	
	RadStd	20	47	86	132	95	168	161	91	72	47	29	18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (59 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page